CORVINUS UNIVERSITY of BUDAPEST

Machine Learning in Practice I
Lecture 8

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Regression

Regression

Regression is optimization in a continuous parameter space

- Data model → feature optimization:
  - we need a parametric combination of the original variables
  - optimize with respect to some condition (c.f. SVM)
- Continuous value estimation:
  - the task can be to predict a continuous variable
  - classification can be based on probability distribution

Regression

task of regression

- we have data pairs (X, Y) → database
- we are looking for a model that $f(x;\omega)=y, \text{ for all } x, y \in X \times Y$
  - we can not consider all possible functions (too large space)
  - fix a functional space (e.g. polynomials, Gaussians, etc.): $f \in F$
  - the variables omega select the actual function

Regression

task of regression

- there is no perfect model! → *philosophical problem …*
  - not enough observation (data)
  - data do not contain all information
  - functional space does not cover all possible effects
- consequence: $f(x;\omega)=y, \text{ for all } x, y \in X \times Y$ can not be satisfied
- what can be achieved: best representation, where the error is the smallest → what is “best”, and what is “error”?

Regression

task of regression – theoretical approach

- not enough information: x→y relation is not unique
- there is a probability to measure a *(data, label)* pair: $\rho(x,y)\in[0,1]$
- for the complete dataset, assuming independent measurements

$$\rho(X,Y)=\prod_{n=1}^{N}\rho(x_n,y_n)$$

- for a given input there can be a lot of outputs: conditional probability (instead of a function) $\rho(Y\mid X)=\dfrac{\rho(X,Y)}{\rho(X)}$

Regression

Bayes theorem

$$p(A\mid B)=\frac{p(A,B)}{p(B)}$$

task of regression – theoretical approach

- data model: $\rho(Y\mid X)\Rightarrow\rho(Y\mid X;\omega)$
- what is the best parameter choice? → *ensemble of worlds*
- define the “distribution of the parameters” using Bayes theorem

$$p(\omega\mid X,Y)=\frac{p(\omega,Y\mid X)}{p(Y\mid X)}=\frac{p(Y\mid\omega,X)\,p(\omega\mid X)}{p(Y\mid X)}$$

- as the function of the parameter → *likelihood function* *(assuming no restriction to possible parameter values)*

$$L(\omega)=p(\omega\mid X,Y)\sim p(Y\mid\omega,X)$$

Regression

task of regression – theoretical approach

- best parameter value when $L(\omega)=p(\omega\,|\,X,Y)$ is maximal *(maximum likelihood):*

$$\frac{\partial L}{\partial \omega_i}(\omega_0)=0$$

- *confidence levels*: $\Omega_p=\{\omega \mid L(\omega)\geq L_p=\text{constant}\} \rightarrow$ hypersurfaces
- a confidence level can be characterized with the probability to be inside the volume $p=P(\omega\in\Omega_p)$
  - *for example: the probability that* $\omega\in\Omega_{0.95}$ *is 95% (according to* $L(\omega)$*)*

Regression

task of regression – theoretical approach

$$N(\mu,C)(y)=\frac{1}{\sqrt{\det(2\pi C)}}e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

- in practice we use **Gaussian** *(central limit theorem)*
  - assume that data points in the database are independent
  - data model gives the expected value
  - with unit correlation matrix → least square method (LSM)

$$\rho(Y\,|\,X;\omega)=\prod_{n=1}^{N}N(f(x_n;\omega),C_n)(y_n)\sim e^{-\frac{1}{2}\sum_{n=1}^{N}(y_n-f(x_n;\omega))C_n^{-1}(y_n-f(x_n;\omega))}\Rightarrow e^{-\frac{1}{2}\chi^2}$$

- the likelihood $L(\omega)\sim e^{-\frac{1}{2}\chi^2(\omega)}$ → not Gaussian (except linear model)

task of regression – theoretical approach

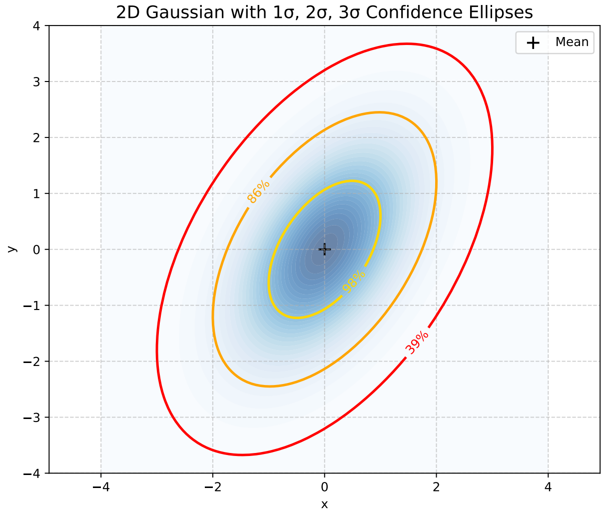
$$N(\mu,C)(y)=\frac{1}{\sqrt{\det(2\pi C)}}e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

- expanding near the maximum likelihood → Gaussian with

$$\chi^2=\chi_0^2+(\omega-\omega_0)R^{-1}(\omega-\omega_0)+\dots$$

- Gaussian confidence levels
  - the confidence surfaces are ellipsoids:
  - confidence levels in **one dimension** $\chi^2-\chi_0^2=n$

sigma-level	1	2	3	4	5
confidence	68.27%	99.54%	99.73%	99.9937%	99.99994%



Regression

task of regression – theoretical approach

$$N(\mu,C)(y)=\frac{1}{\sqrt{\det(2\pi C)}}e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

- Gaussian confidence levels and chi-squared distribution

$$p_d(\chi^2)=\frac{1}{2^{d/2}\Gamma(d/2)}(\chi^2)^{\frac{d}{2}-1}e^{-\frac{1}{2}\chi^2}d\chi^2$$

- probability to be inside $\chi^2<r^2$ with the incomplete Gamma function

$$P(\chi^2<r^2)=\int_0^{r^2}d\chi^2 p_d(\chi^2)=\frac{\gamma(d/2,r^2/2)}{\Gamma(d/2)}$$

$$\gamma(a,x)=\int_0^x x^{a-1}e^{-x}dx$$

$$\Gamma(a)=\gamma(a,\infty)$$

- in Python: `scipy.special.gammaincc`

$$\frac{1}{\sqrt{\det(2\pi C)}}e^{-\frac{1}{2}(y-\mu)^T C^{-1}(y-\mu)}d^d y=\frac{1}{(2\pi)^{d/2}}e^{-\frac{1}{2}z^T z}d^d z=\frac{1}{2^{d/2}\Gamma(d/2)}(\chi^2)^{\frac{d}{2}-1}e^{-\frac{1}{2}\chi^2}d\chi^2$$

Regression

$$N(\mu, C)(y) = \frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

task of regression – theoretical approach

- calculation with Gaussian distribution: practical rules
  - assume *y* is Gaussian
  - introduce zero-mean variables $\xi = y - \mu$
  - expected values: $\mathrm{E}(\xi) = 0, \quad E(\xi_i \xi_j) = C_{ij} \quad \Rightarrow \quad E(\xi\xi) = C$
  - higher expected values are product of pairs!
    e.g.: $E(\xi_i \xi_j \xi_k \xi_l) = C_{ij} C_{kl} + C_{ik} C_{jl} + C_{il} C_{jk}$

Regression

task of regression – theoretical approach

$$N(\mu,C)(y)=\frac{1}{\sqrt{\det(2\pi C)}}e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

- hypothesis checking with chi-squared distribution:
  - assume Gaussian hypothesis and the data model are correct
  - treat the labels as independent random variables $y_n - f(x_n,\omega) \Rightarrow \xi_n$
  - compute expected value of chi-squared

$$\mathrm{E}(\chi^2)=\mathrm{E}\left(\sum_{n=1}^{N}\xi_n C_n^{-1}\xi_n\right)=\sum_{n=1}^{N}\sum_{i,j=1}^{D}(C_n^{-1})_{ij}\,\mathrm{E}(\xi_{ni}\xi_{nj})=N$$

- *chi-squared value over degrees of freedom is one*

Regression
Linear regression

Linear regression

simplest relation: linear function

- data are $(X,Y)=\{(x_n,y_n)\,|\,x_n\in\mathbb{R}^{N_x},\ y_n\in\mathbb{R}^{N_y},\ n=1,2,\ldots N\}$
- assume linear relation: $f(x;a,b)=ax+b$
- coefficient *a* is an matrix, *b* is a vector $[a]\Rightarrow N_x\times N_y,\ [b]\Rightarrow N_y$
- task to find the value (distribution or likelihood) of the coefficients
- simplest case, in lot of cases very powerful

Linear regression

one-dimensional case

- simplest: 1D linear regression, diagonal correlation matrix
- problem: data and labels are real numbers

$$(X,Y)=\{(x_n,y_n)\,|\,x_n,y_n\in\mathbb{R},\ n=1,2,\ldots N\}$$

- assume there is a linear relation between the two: $f(x;a,b)=ax+b$
- chi-squared function $\chi^2=\sum_{n=1}^{N}\frac{(ax_n+b-y_n)^2}{\sigma_n^2}$
- likelihood is Gaussian in a,b: $L(\omega)\sim e^{-\frac{1}{2}\chi^2(\omega)}=e^{\sum_{n=1}^{N}\frac{(ax_n+b-y_n)^2}{2\sigma_n^2}}$

1D linear case

- computation: introduce new notations
  - sum of inverse variances: $\sigma^2 = \frac{1}{N} \sum_{n=1}^{N} \frac{1}{\sigma_n^2}$
  - averaging: $\langle u \rangle = \frac{\sum_n u_n / \sigma_n^2}{\sum_n 1/\sigma_n^2} \Rightarrow \frac{1}{N} \sum_{n=1}^{N} u_n$
  - correlation: $C_{uv} = \langle uv \rangle - \langle u \rangle \langle v \rangle = \langle (u - \langle u \rangle)(v - \langle v \rangle) \rangle$
  - $\bar{b} = b + a\langle x \rangle - \langle y \rangle \;\Rightarrow\; y = \langle y \rangle + a(x - \langle x \rangle) + \bar{b}$

Linear regression

1D linear case

- computation: $\chi^2=\left(a-C_{yx}C_{xx}^{-1}\right)\frac{NC_{xx}}{\sigma^2}\left(a-C_{yx}C_{xx}^{-1}\right)^T+\frac{N\overline{b^2}}{N}+\text{constant}$
  - a and $\bar{b}$ are independent Gaussian variables

$$\mu_a=C_{yx}C_{xx}^{-1},\quad C_a^{-1}=\frac{N\,C_{xx}}{\sigma^2},\quad C_{\bar{b}}^{-1}=\frac{N}{\sigma^2}$$

  - the distribution of the variable y=ax+b (with fixed x) is also Gaussian

$$\mu_y=\langle y\rangle+\mu_a\left(x-\langle x\rangle\right),\quad C_y=\left(x-\langle x\rangle\right)C_a\left(x-\langle x\rangle\right)+C_{\bar{b}}$$

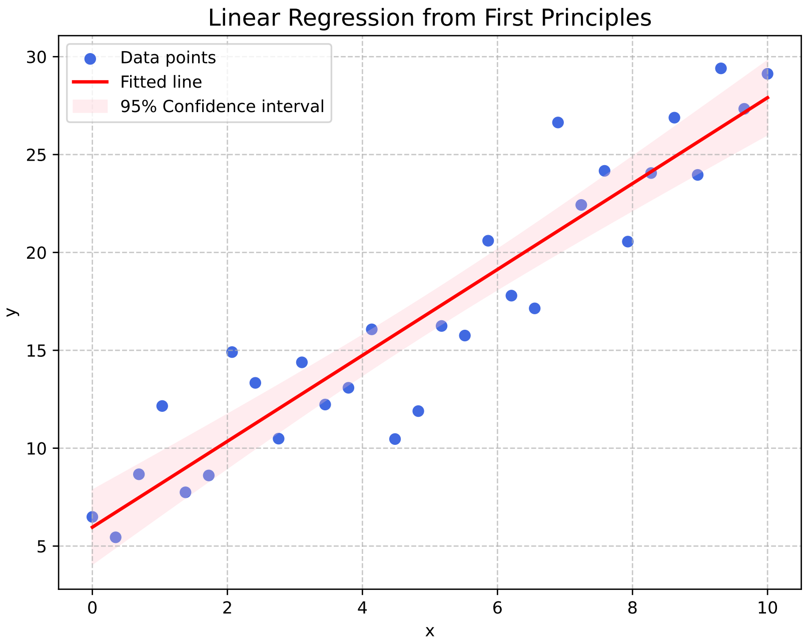
1D linear case

- interpretation:
  - best slope value and variance

$$a = C_{yx} C_{xx}^{-1}, \quad \sigma_a^2 = \frac{1}{C_{xx} \sum_n 1/\sigma_n^2} \Rightarrow \frac{\sigma^2}{\sum_n (x_n - \bar{x})}$$

  - chi-squared expected value

$$\frac{1}{N} \sum_n \frac{(y_n - a x_n - b)^2}{\sigma_n^2} = 0.81$$



Linear regression

higher dimensional linear case

- data are $(X,Y)=\{(x_n,y_n)\,|\,x_n\in\mathbb{R}^{N_x},\ y_n\in\mathbb{R}^{N_y},\ n=1,2,\ldots N\}$
- assume linear relation: $f(x;a,b)=ax+b$
- coefficient *a* is an matrix, *b* is a vector $[a]\Rightarrow N_x\times N_y,\ [b]\Rightarrow N_y$
- calculation and formulae remain the same

$$a=C_{yx}C_{xx}^{-1},\quad C_a^{-1}=\frac{N}{\sigma^2}C_{xx}$$

Linear regression

higher dimensional linear case

- example: $N_x = N_y = 2, N_{data} = 100,$

$$a_{true} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad b_{true} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- then the result is

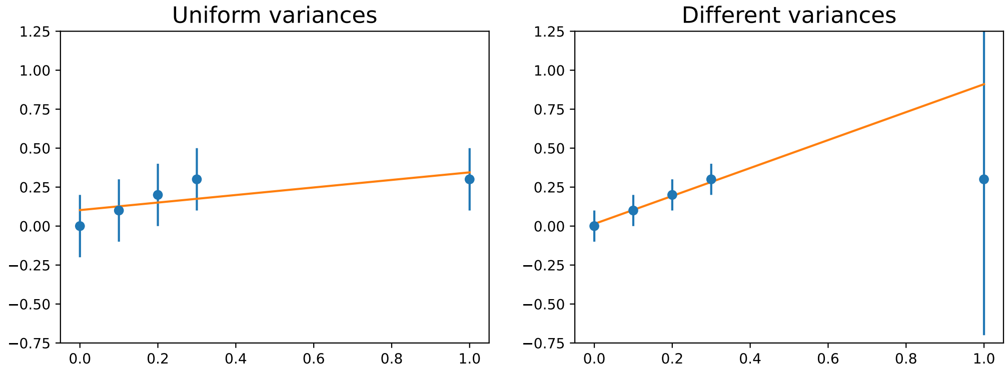
$$a_{data} = \begin{pmatrix} 0.99 & 2.03 \\ 3.06 & 3.96 \end{pmatrix}, \quad b_{data} = \begin{pmatrix} 0.98 \\ -0.98 \end{pmatrix}$$

```
# linear regression in 2D
Ndata = 100

a = np.array([[1, 2],[3, 4]])
b = np.array([1,-1])
x = np.random.uniform(0,1, size=(2, Ndata))
noise = np.random.normal(0,0.1, size=(2, Ndata))
y = a@x + noise + b[:,None]

barx = np.mean(x, axis=1)
bary = np.mean(y, axis=1)
Cxx = (x-barx[:,None])@(x-barx[:,None]).T/2
Cyx = (y-bary[:,None])@(x-barx[:,None]).T/2

a1 = Cyx@np.linalg.inv(Cxx)
b1 = bary - a1@barx
omega1, b1
```

Linear regression

role of the individual variances

- effect of different variances → weight importance of points



- variance
  - measurement (tedious)
  - theoretical considerations (e.g. relative error is $\sigma \sim \sqrt{N}$)
  - user expectations (e.g. some points are important)

Linear regression

role of the individual variances

- correlation matrix from measurements:
  - assume that the data are i.i.d. normal $p(x) \sim \prod_{n=1}^{N} e^{-\frac{1}{2}(x_n-\mu)C^{-1}(x_n-\mu)}$
  - distribution of $\overline{x} = \frac{1}{N}\sum_{n=1}^{N} x_n$ is normal with $\mathrm{E}(\overline{x}) = \mu$ → estimate for mean
  - mean of $\overline{C_{ij}} = \frac{1}{N-1}\sum_{n=1}^{N}(x_n-\overline{x})_i(x_n-\overline{x})_j$ → $\mathrm{E}(\overline{C_{ij}}) = C_{ij}$ → variance

END OF LECTURE 8